

Chapter 21: Optical Properties

ISSUES TO ADDRESS...

- What phenomena occur when light is shined on a material?
- What determines the characteristic colors of materials?
- Why are some materials transparent and others are translucent or opaque?

Optical Properties

Light has both particulate and wavelike characteristics

- **Photon** - a quantum unit of light

$$E = h\nu = \frac{hc}{\lambda}$$



E = energy of a photon



λ = wavelength of radiation



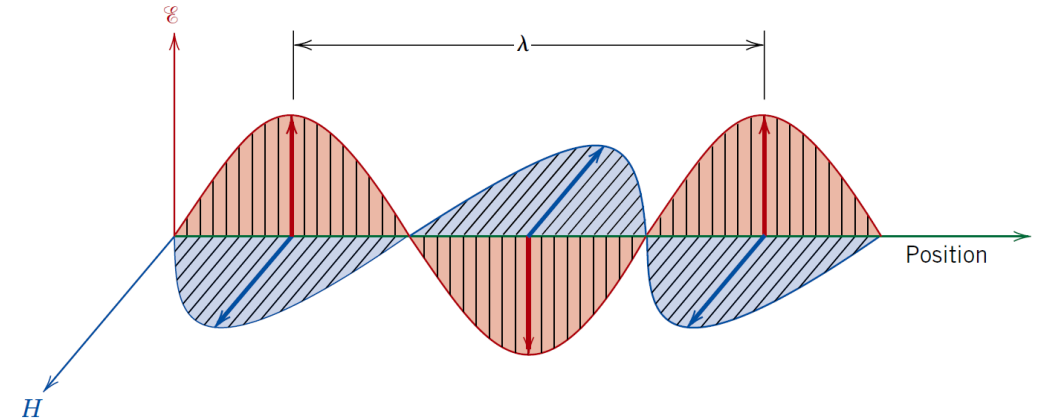
ν = frequency of radiation

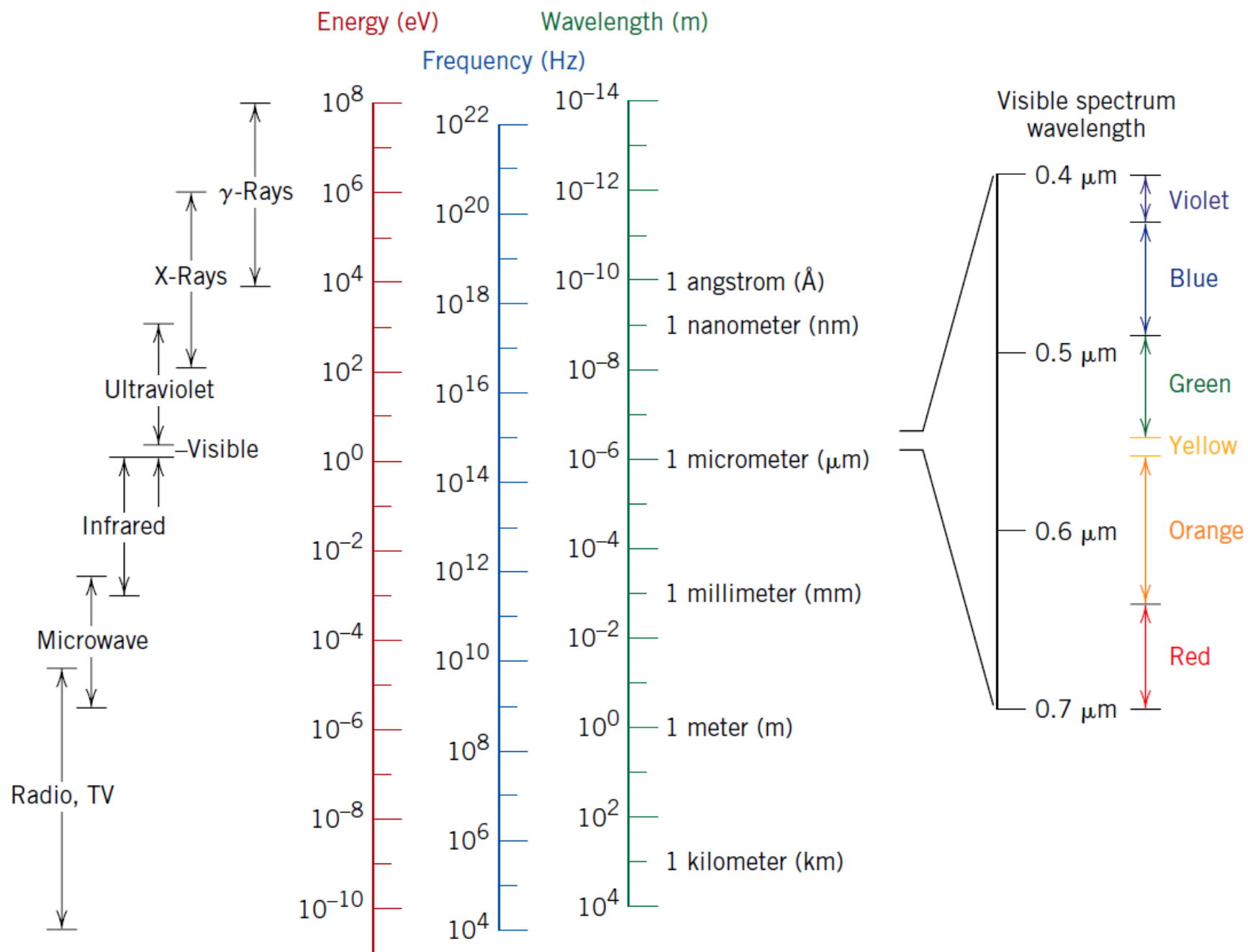


h = Planck's constant ($6.62 \times 10^{-34} \text{ J} \cdot \text{s}$)



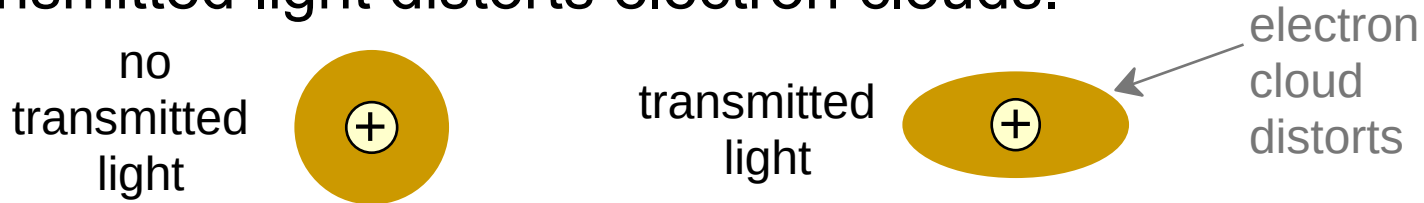
c = speed of light in a vacuum ($3.00 \times 10^8 \text{ m/s}$)





Refraction

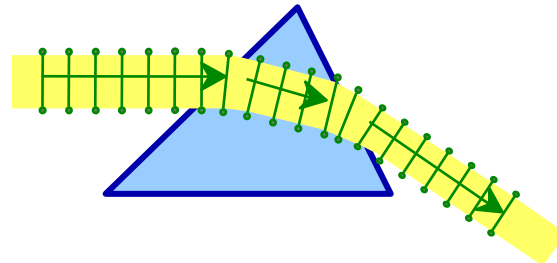
- Transmitted light distorts electron clouds.



- The velocity of light in a material is lower than in a vacuum.

$$n = \text{index of refraction} \equiv \frac{c \text{ (velocity of light in vacuum)}}{v \text{ (velocity of light in medium)}}$$

- Adding large ions (e.g., lead) to glass decreases the speed of light in the glass.
- Light can be “bent” as it passes through a transparent prism

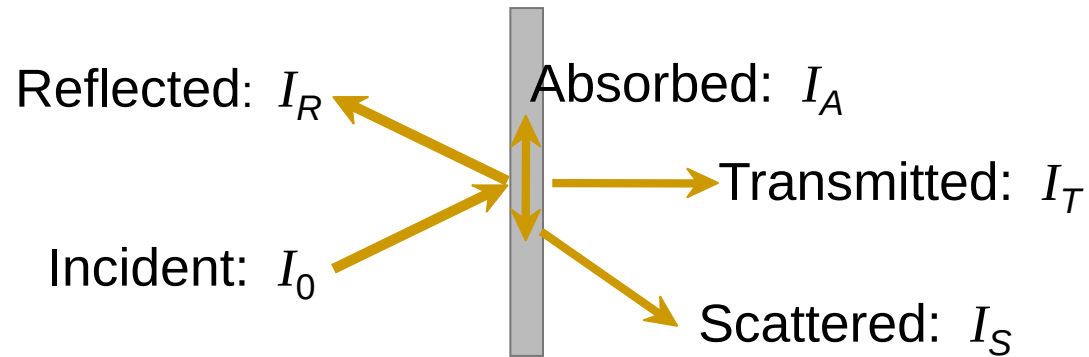


Material	n
Typical glasses ca.	1.5 -1.7
Plastics	1.3 -1.6
PbO (Litharge)	2.67
Diamond	2.41

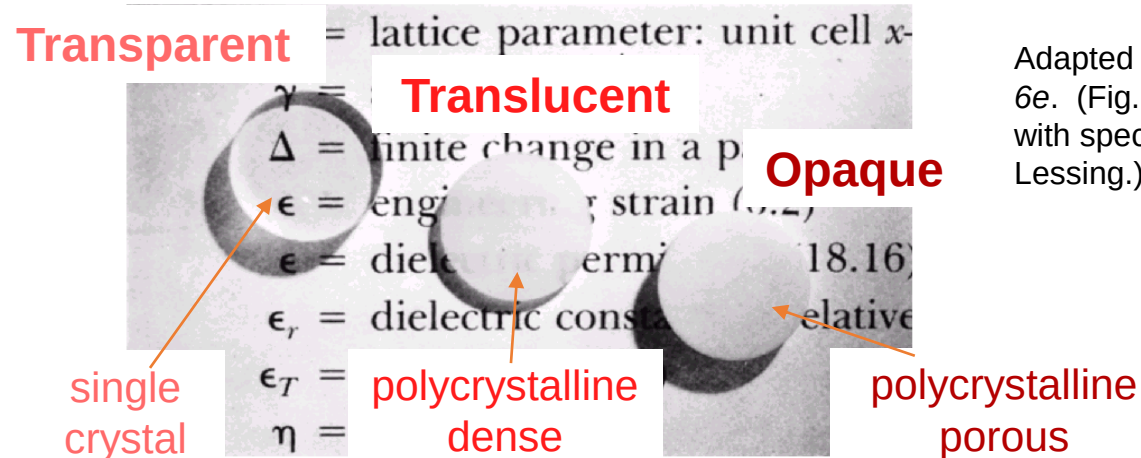
Selected values from Table 21.1,
Callister & Rethwisch 8e.

Light Interactions with Solids

- Incident light is reflected, absorbed, scattered, and/or transmitted: $I_0 = I_T + I_A + I_R + I_S$



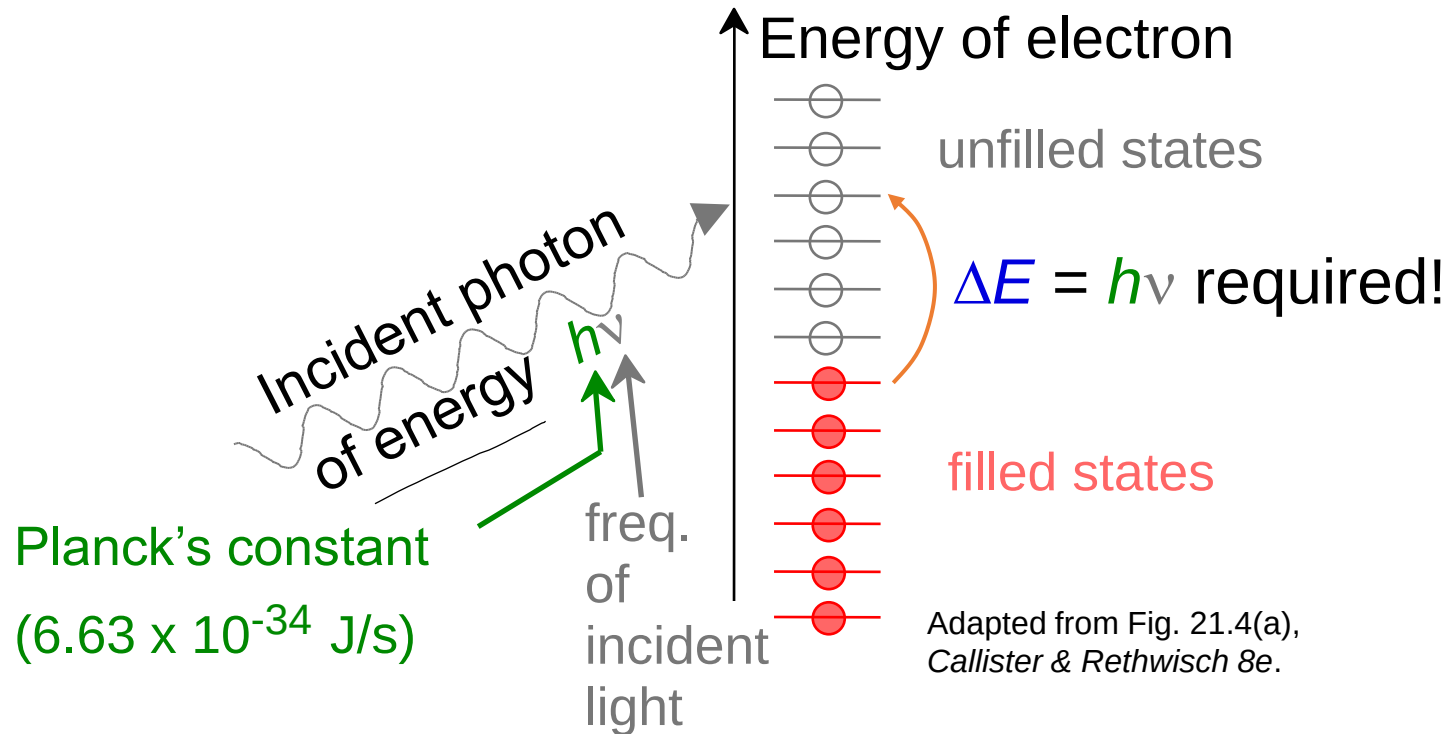
- Optical classification of materials:



Adapted from Fig. 21.10, *Callister 6e*. (Fig. 21.10 is by J. Telford, with specimen preparation by P.A. Lessing.)

Optical Properties of Metals: Absorption

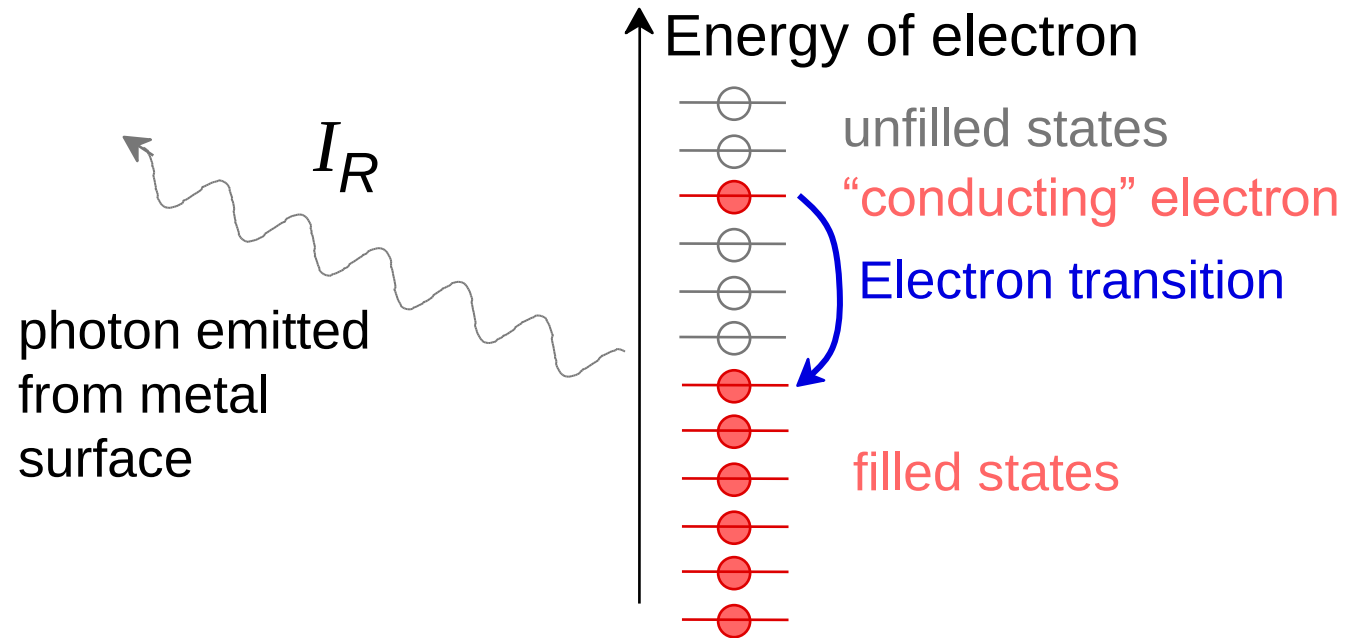
- **Absorption** of photons by electron transitions:



- Unfilled electron states are adjacent to filled states
- Near-surface electrons absorb visible light.

Reflection of Light for Metals

- **Electron transition** from an excited state produces a photon.



Adapted from Fig. 21.4(b),
Callister & Rethwisch 8e.

Reflection of Light for Metals (cont.)

- **Reflectivity** = I_R/I_0 is between 0.90 and 0.95.
- Metal surfaces appear shiny
- Most of absorbed light is reflected at the same wavelength
- Small fraction of light may be absorbed
- Color of reflected light depends on wavelength distribution
 - Example: The metals copper and gold absorb light in blue and green => reflected light has gold color

Reflectivity of Nonmetals

- For normal incidence and light passing into a solid having an index of refraction n :

$$R = \text{reflectivity} = \left(\frac{n-1}{n+1} \right)^2$$

- Example: For Diamond $n = 2.41$

$$R = \left(\frac{2.41-1}{2.41+1} \right)^2 = 0.17$$

$\therefore$ 17% of light is reflected

Selected Light Absorption in Semiconductors

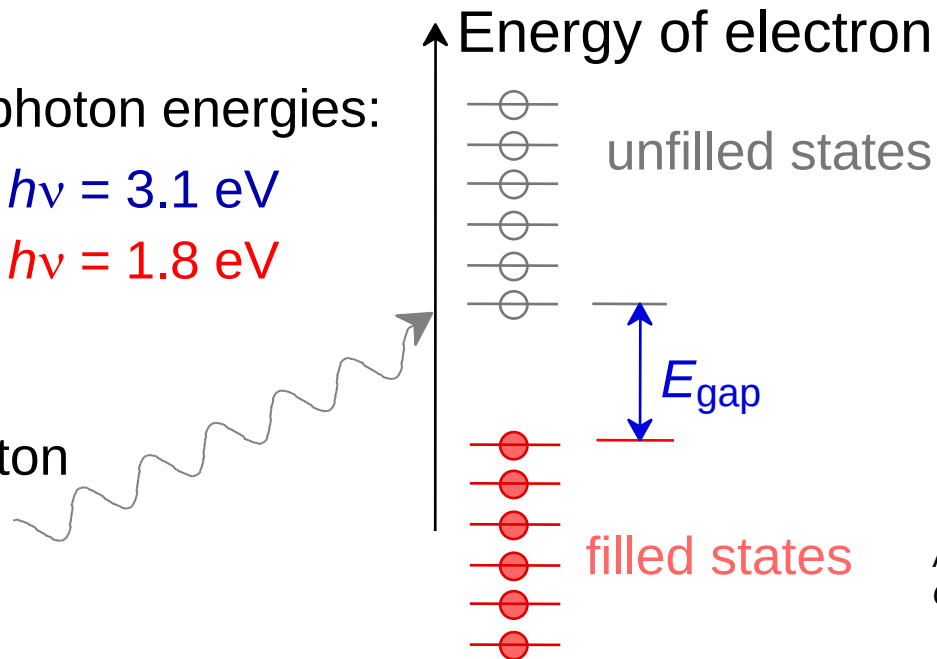
Absorption of light of frequency ν by electron transition occurs if $h\nu > E_{\text{gap}}$

Examples of photon energies:

blue light: $h\nu = 3.1 \text{ eV}$

red light: $h\nu = 1.8 \text{ eV}$

incident photon
energy $h\nu$

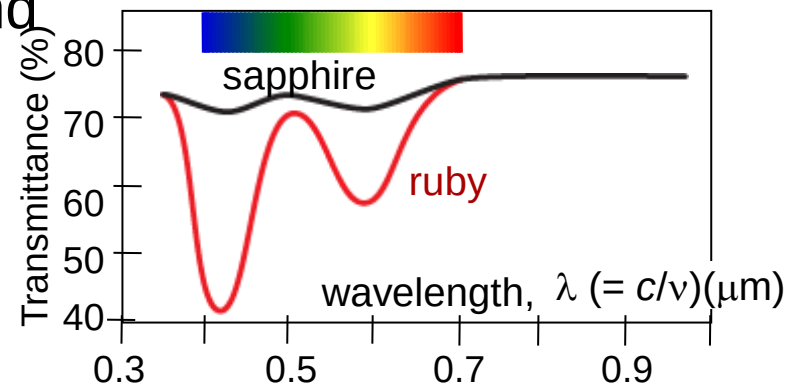


Adapted from Fig. 21.5(a),
Callister & Rethwisch 8e.

- If $E_{\text{gap}} < 1.8 \text{ eV}$, all light absorbed; material is opaque (e.g., Si, GaAs)
- If $1.8 \text{ eV} < E_{\text{gap}} < 3.1 \text{ eV}$, partial light absorption; material is colored
- If $E_{\text{gap}} > 3.1 \text{ eV}$, no light absorption; material is transparent and colorless (e.g., diamond)

Color of Nonmetals

- Color determined by the distribution of wavelengths:
 - transmitted light
 - re-emitted light from electron transitions
- Example 1: Cadmium Sulfide (CdS), $E_g = 2.4$ eV
 - absorbs higher energy visible light (blue, violet)
 - color results from red/orange/yellow light that is transmitted
- Example 2: Ruby = Sapphire (Al_2O_3) + (0.5 to 2) at% Cr_2O_3
 - Sapphire is transparent and colorless ($E_g > 3.1$ eV)
 - adding Cr_2O_3 :
 - alters the band gap
 - blue and orange/yellow/green light is absorbed
 - red light is transmitted
 - Result: Ruby is deep red in color



Adapted from Fig. 21.9, Callister & Rethwisch 8e.
(Fig. 21.9 adapted from "The Optical Properties of Materials" by A. Javan, *Scientific American*, 1967.)

TRANSMITTED LIGHT: REFRACTION

- Transmitted light distorts electron clouds.

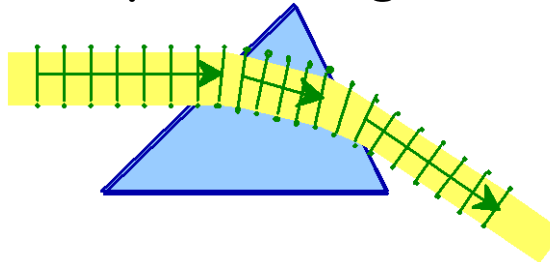


- Result 1: Light is slower in a material vs vacuum.

Index of refraction (n) = $\frac{\text{speed of light in a vacuum}}{\text{speed of light in a material}}$

--Adding large, heavy ions (e.g., **lead**) can decrease the speed of light.

--Light can be "bent"



Material	n
Lead glass	2.1
Silica glass	1.46
Soda-lime glass	1.51
Quartz	1.55
Plexiglas	1.49
Polypropylene	1.49

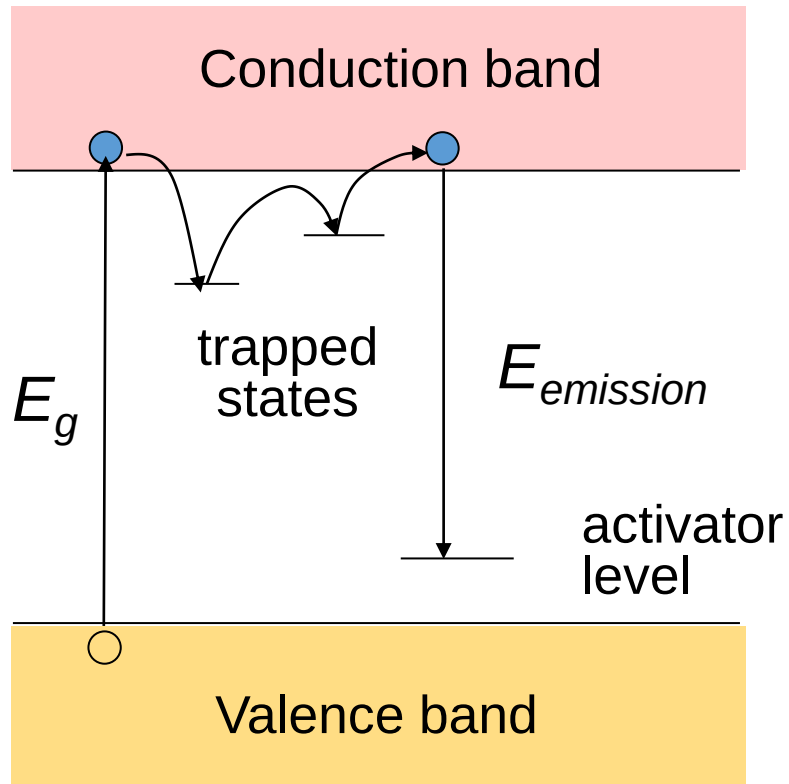
Selected values from Table 21.1,
Callister 6e.

- Result 2: Intensity of transmitted light decreases with distance traveled (thick pieces less transparent!)

Applications of Optical Phenomena

Luminescence

- **Luminescence** – reemission of light by a material
 - Material absorbs light at one frequency and reemits it at another (lower) frequency.
 - Trapped (donor/acceptor) states introduced by impurities/defects

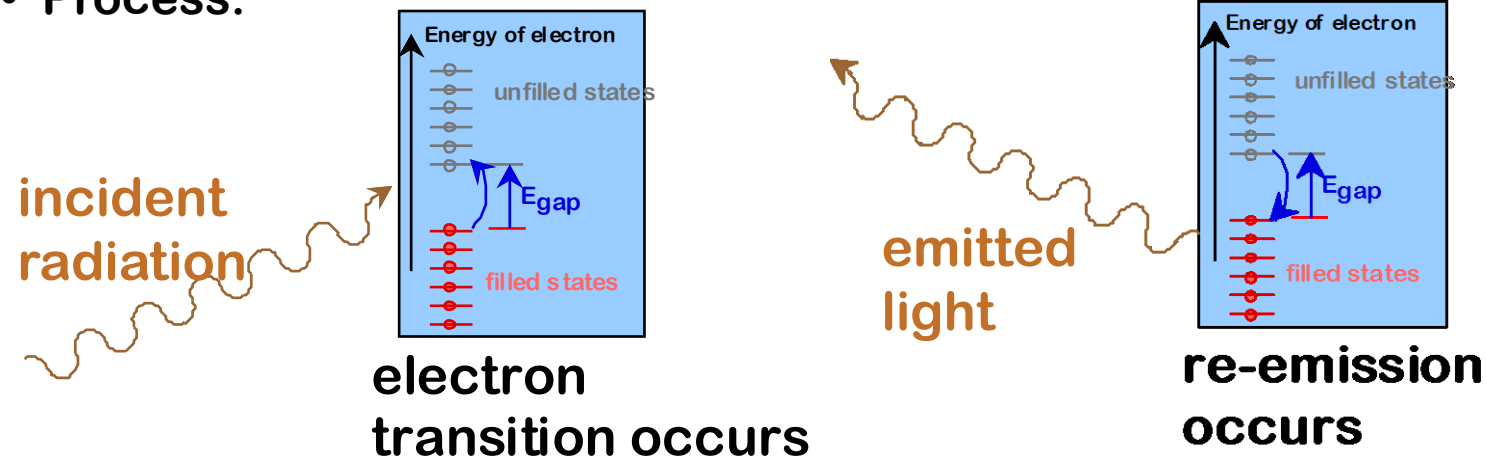


- If residence time in trapped state is relatively long ($> 10^{-8}$ s)
-- phosphorescence
- For short residence times ($< 10^{-8}$ s)
-- fluorescence

Example: Toys that glow in the dark. Charge toys by exposing them to light. Reemission of light over time—phosphorescence

APPLICATION: LUMINESCENCE

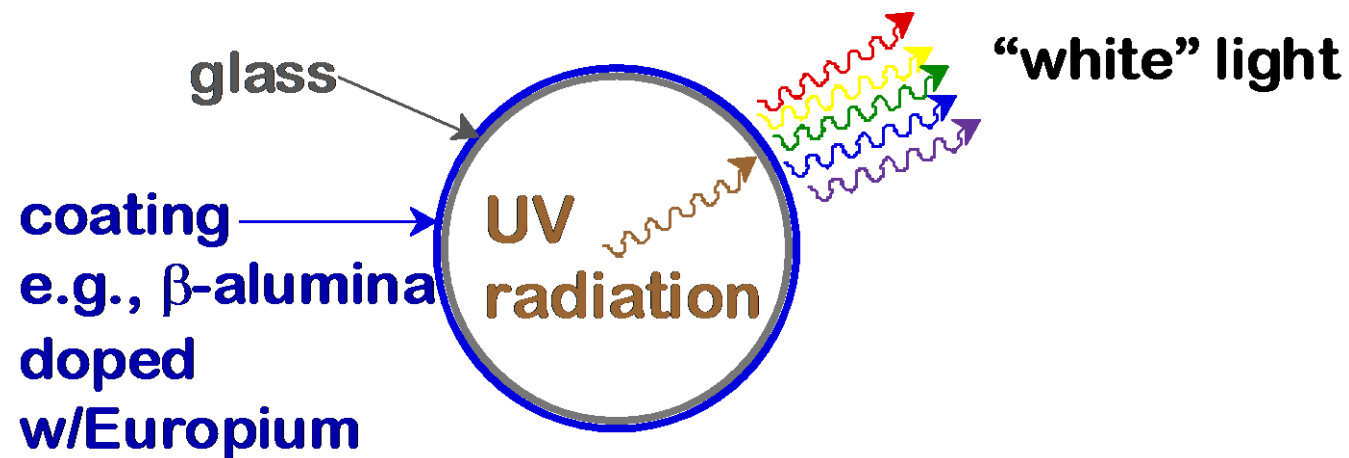
- **Process:**



Adapted from Fig. 21.5(a), *Callister 6e*.

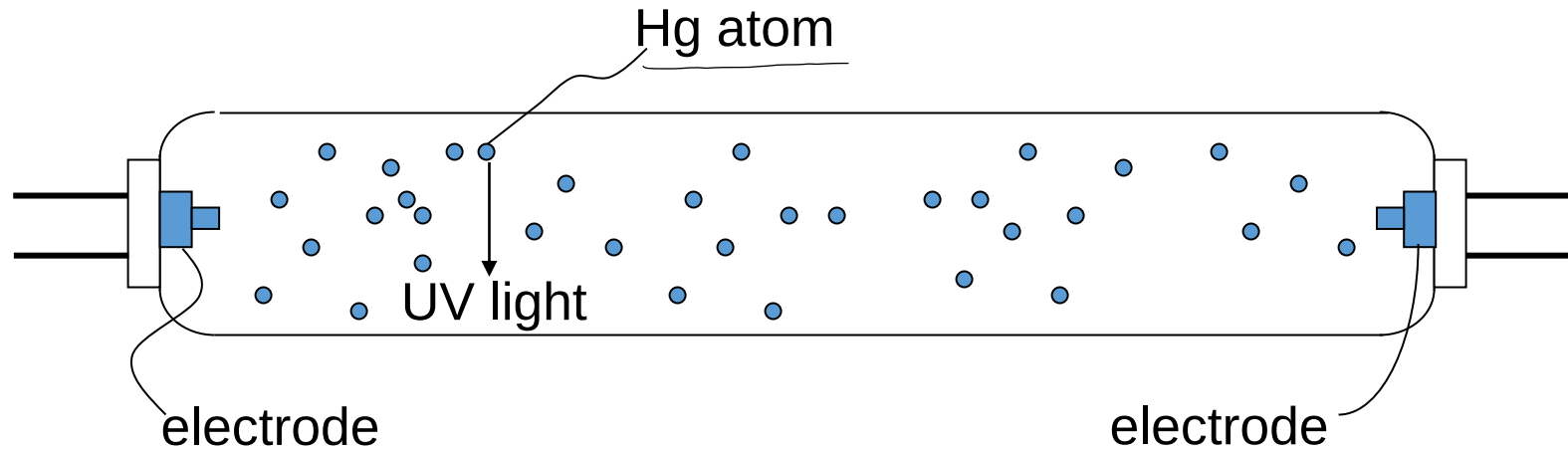
Adapted from Fig. 21.5(a), *Callister 6e*.

- **Ex: fluorescent lamps**



Photoluminescence

When photons impinge on a material which in turn re-emits light of a lower energy



- Arc between electrodes excites electrons in mercury atoms in the lamp to higher energy levels.
- As electron falls back into their ground states, UV light is emitted (e.g., suntan lamp).
- Inside surface of tube lined with material that absorbs UV and reemits visible light
 - For example, $\text{Ca}_{10}\text{F}_2\text{P}_6\text{O}_{24}$ with 20% of F^- replaced by Cl^-
- Adjust color by doping with metal cations
 - Sb^{3+} blue
 - Mn^{2+} orange-red

- Electroluminescence – *light emission as a consequence of an applied voltage or electric field.*
- Cathodoluminescence – *light emission from a substance that has been showered by electrons of higher energy.*

The LASER

- The laser generates light waves that are in phase (coherent) and that travel parallel to one another
 - LASER
 - Light
 - Amplification by
 - Stimulated
 - Emission of
 - Radiation
- Operation of laser involves a population inversion of energy states process

Population Inversion

- More electrons in excited energy states than in ground states

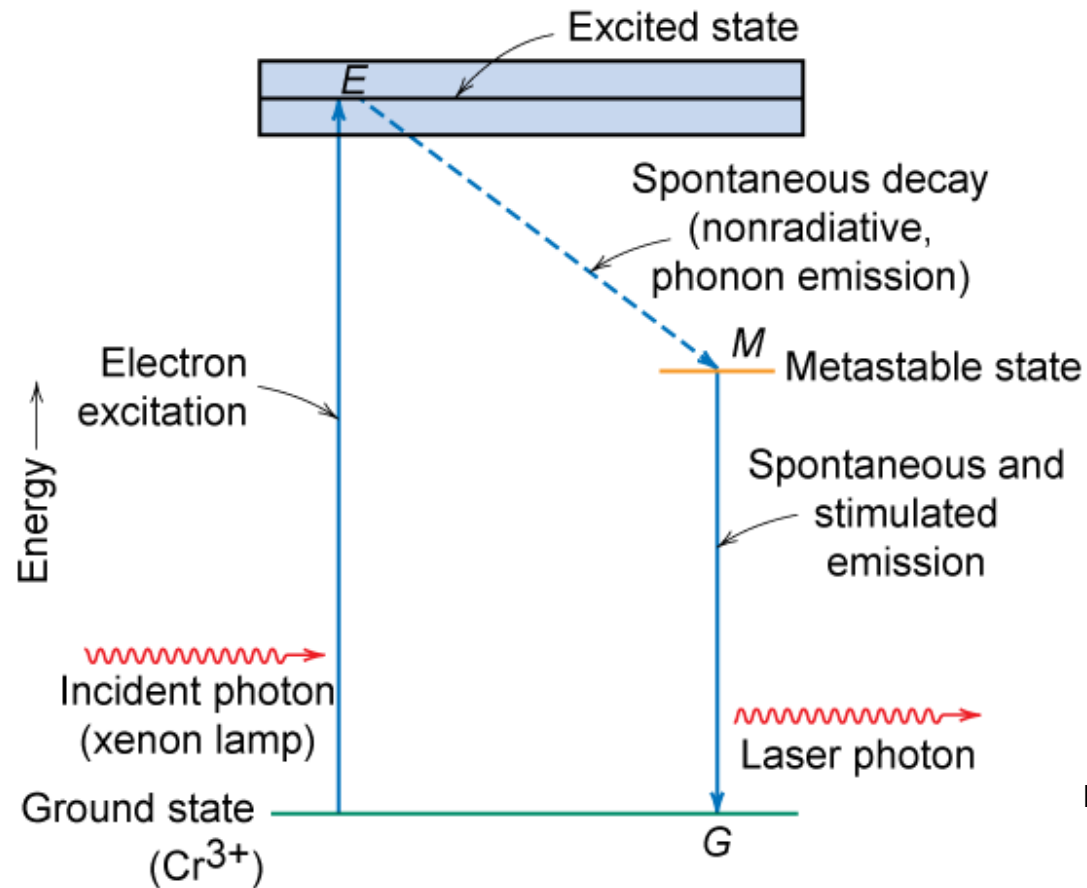
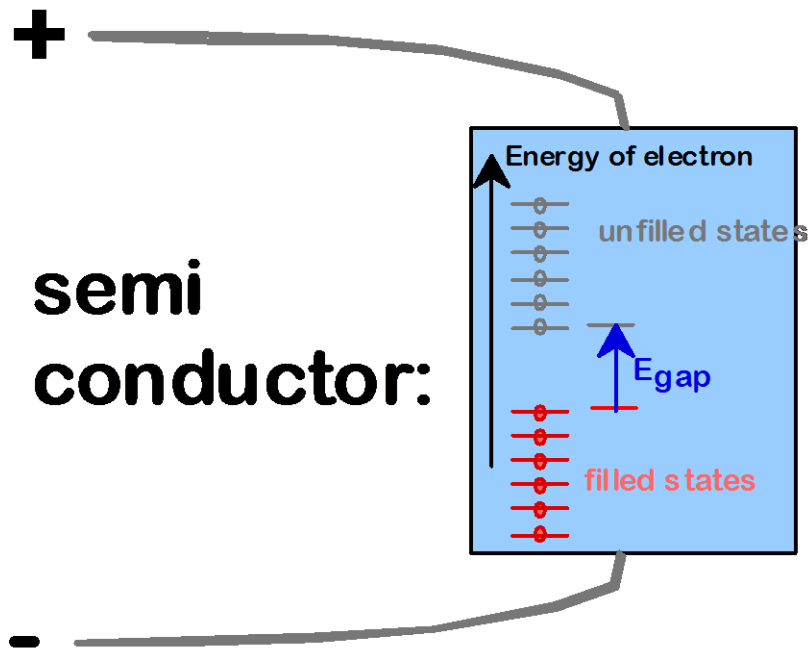


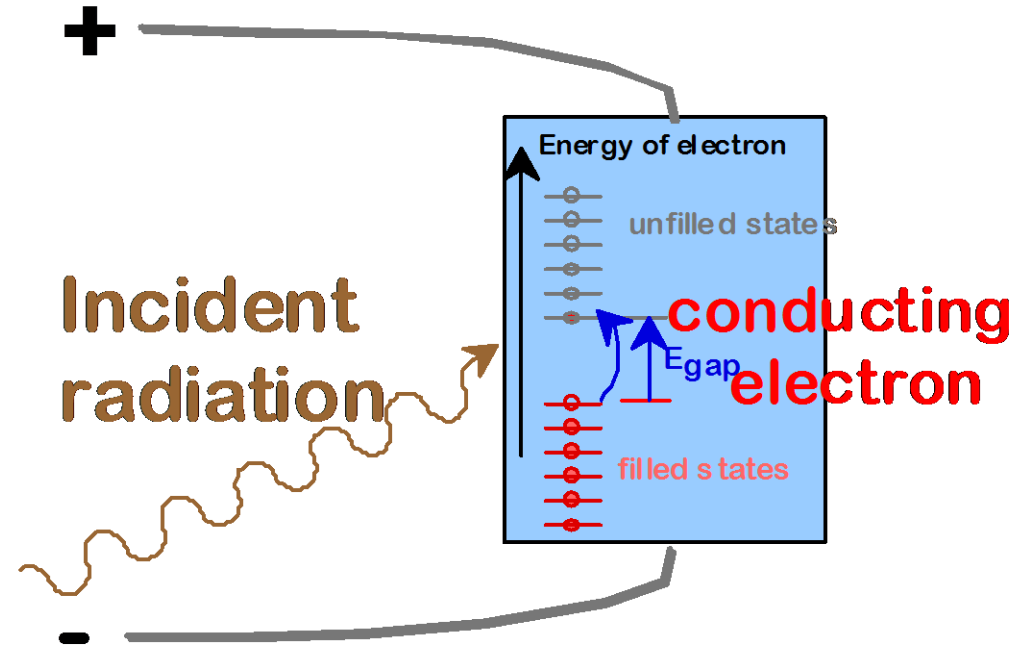
Fig. 21.14, Callister & Rethwisch 8e.

APPLICATION: PHOTOCONDUCTIVITY

- Description:



**A. No incident radiation:
little current flow**

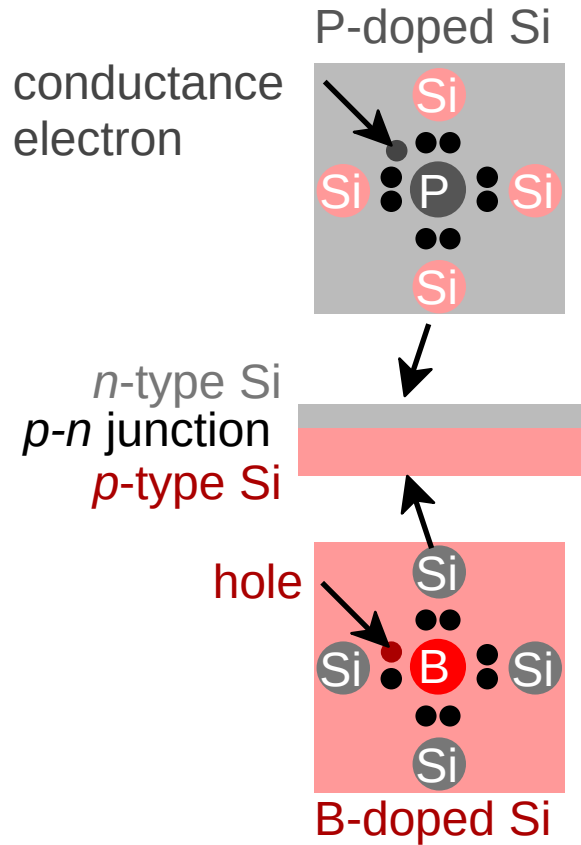


**B. Incident radiation:
increased current flow**

- Ex: Photodetector (Cadmium sulfide)

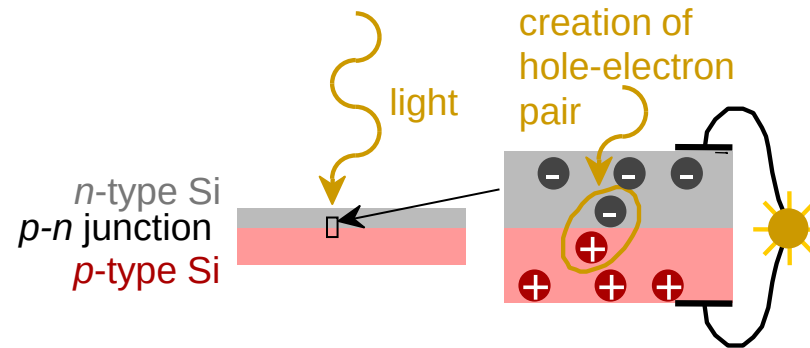
Other Applications - Solar Cells

- p - n junction:

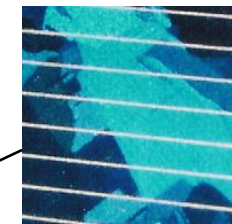


- Operation:

- incident photon of light produces elec.-hole pair.
- typical potential of 0.5 V produced across junction
- current increases w/light intensity.



- Solar powered weather station:

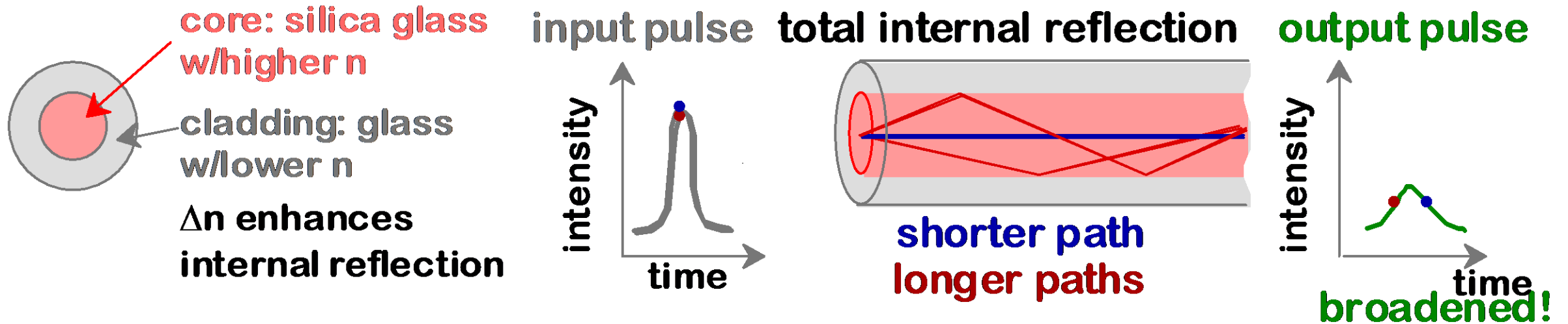


polycrystalline Si

Los Alamos High School weather station (photo courtesy P.M. Anderson)

APPLICATION: FIBER OPTICS

- Design with stepped index of refraction (n):



Adapted from Fig. 21.19, *Callister 6e*. (Fig. 21.19 adapted from S.R. Nagel, *IEEE Communications Magazine*, Vol. 25, No. 4, p. 34, 1987.)

SUMMARY

- Light radiation impinging on a material may be reflected from, absorbed within, and/or transmitted through
- Light transmission characteristics:
 - transparent, translucent, opaque
- Optical properties of metals:
 - opaque and highly reflective due to electron energy band structure.
- Optical properties of non-Metals:
 - for $E_{gap} < 1.8$ eV, absorption of all wavelengths of light radiation
 - for $E_{gap} > 3.1$ eV, no absorption of visible light radiation
 - for $1.8 \text{ eV} < E_{gap} < 3.1$ eV, absorption of some range of light radiation wavelengths
 - color determined by wavelength distribution of transmitted light
- Other important optical applications/devices:
 - luminescence, photoconductivity, light-emitting diodes, solar cells, lasers